

# ABHINASH PRADHAN

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## PROFESSIONAL SUMMARY

Computer Science undergraduate completed an AI internship at OCAC (Govt. of Odisha) building ML pipelines on real financial data, reducing manual analysis time by an estimated 40%. Trained a plant disease classifier achieving 99.29% validation accuracy across 38 classes. Built and deployed a real-time chat app with Firebase Auth and Socket.io.

## TECHNICAL SKILLS

**Languages:** Java, Python, JavaScript

**ML & Data:** Scikit-learn, Pandas, NumPy, Matplotlib, PyTorch, Computer Vision, Transfer Learning, Classification, Regression

**Backend:** Node.js, Express.js, Socket.io, REST APIs

**Frontend:** React.js, HTML5, CSS3, Tailwind CSS

**Database:** MongoDB

**Tools:** Git, GitHub, VS Code, Docker, Vercel

**Concepts:** JWT Authentication, Data Structures & Algorithms, DBMS, OOP

## PROFESSIONAL EXPERIENCE

### Artificial Intelligence Intern| OCAC

May 2025 - July 2025

Bhubaneswar, India

- Built an invoice payment prediction system in Python using Scikit-learn, Pandas, and NumPy — achieving **F1 score of 0.82** on held-out test set with 80/20 train-test split across 1,000+ real financial records
- Reduced model error by **~23%** through feature engineering — encoded 5+ categorical variables, handled missing values, and normalized skewed distributions on structured invoice data
- Built end-to-end ML pipeline — CSV ingestion → Pandas preprocessing → Scikit-learn model training → prediction output — reducing manual financial analysis time by an estimated **40%**

## PROJECTS

### Plant Disease Prediction

Oct 2025 - Jan 2026

Python, PyTorch, EfficientNet-B0, ResNet-50, Computer Vision

- Fine-tuned EfficientNet-B0 and ResNet-50 using PyTorch transfer learning on the PlantVillage dataset — classifying **38 plant disease classes** with **99.29% val accuracy** across 70/15/15 train-val-test split over 15 epochs
- Built end-to-end image classification pipeline — data augmentation (random crop, flip, normalization) → model training with Adam optimizer → exported deployable EfficientNet model as a .pkl file for real-world leaf disease prediction from images

### Chatify – Real-Time Chat Application | (Live)

Jan 2026 - Apr 2026

React.js, Node.js, Express.js, MongoDB, Socket.io, Firebase Auth, Google OAuth

- Built real-time messaging with Socket.io supporting **50+ concurrent connections** with event-driven architecture and live message delivery
- Implemented hybrid auth using Firebase Google OAuth and JWT — enabling one-click sign-in with persistent sessions and role-based access control
- Developed core features including one-to-one messaging, user presence indicators, chat history persistence in MongoDB, and responsive UI using Tailwind CSS

### ResuMatch – AI Resume Analyzer | (Live)

Apr 2026 - May 2026

React.js, Node.js, Express.js, MongoDB, Firebase Auth, Groq API, Gemini API, OpenAI API

- Built AI resume analyzer scoring resume-to-job match (0–100%) with missing keywords and suggestions using a 3-tier fallback — Groq → Gemini → OpenAI — ensuring zero downtime when any model hits rate limits
- Implemented PDF parsing pipeline with pdf-parse, structured JSON prompt engineering, and per-user analysis history stored in MongoDB Atlas
- Integrated Firebase Google OAuth and Email/Password auth with protected routes and deployed full-stack app on Render

## CERTIFICATIONS & ACHIEVEMENTS

- Deloitte Virtual Internship Program (Data Analytics)** – Credential
- NPTEL Elite** – Industry 4.0 & IoT (IIT Kharagpur) and Affective Computing (IIIT Delhi)
- MongoDB CRUD Operations Certification** – Credential
- Moniba Technology & Innovations** – Programming with Python (July 2024) – Credential

## EDUCATION

### B. Tech in Computer Science and Engineering

GIFT Autonomous, Bhubaneswar | CGPA: 7.9/10 | 2023 – Present

### Intermediate (12th)

Godavarish Higher Secondary School, Banpur (CHSE) | 2021 – 2023